INCH-POUND

MIL-O-55310/7E
15 March 1994
SUPERSEDING
MIL-O-55310/7D
4 October 1984

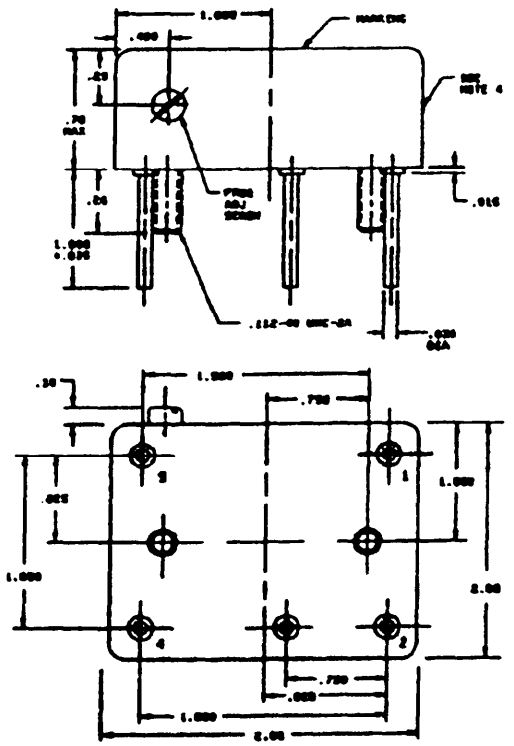
MILITARY SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 3 (TEMPERATURE COMPENSATED (TCXO)), 10 THROUGH 90 MHz, SOLDER SEAL, SINE WAVE

Inactive for new design

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310



Pin number	Function
1	B+
2	N/C
3	RF output
4	GND
5	B-

Inches	mm
.015	0.38
.030	0.76
.035	0.89
.10	2.5
.112	2.84
.26	6.6
.29	7.4
.400	10.16
.70	17.8
.750	19.05
.825	20.96
1.000	25.40
1.500	38.10
1.650	41.91
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Pin number and function to be displayed on side of oscillator.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Solder (nonhermetic).

Weight: 3 ounces maximum.

Construction technology: Discrete construction.

Components: In accordance with MIL-STD-454.

Calibration: Manufacturer and user nominal frequency calibrated.

Screening: In accordance with MIL-O-55310, class B.

Temperature:

Operating: -55°C to +85°C.

Storage: -62°C to +125°C.

Oscillator load: 50 ohms resistance.

Output waveform: Sine wave.

Supply voltage: +12 V dc ±0.25 V dc.

Input current: 25 mA maximum.

Output frequency: Frequency as designated at time of acquisition; see table I.

Initial accuracy at reference temperature: ±1 ppm at +23°C ±3°C.

Frequency adjustment (mechanical; adjustable after the cover is on): A tuning adjustment for aging shall have a range of ±2.5 ppm. Resolution shall be ±0.1 ppm maximum. Grounding the tuning screw shall not damage the circuit.

Output power: See table I.

Harmonics and subharmonics distortion: See table I.

Initial frequency-temperature accuracy (one-half temperature cycle): See table I. Measurements taken at 10 equispaced increments over the specified operating temperature range.

Frequency-voltage tolerance: See table I for a ±0.25 volt change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

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TABLE I. Dash numbers and operating characteristics.

Dash number	Output frequency range	Output power	Output distortion (max)	Frequency-temperature accuracy	Frequency-voltage tolerance
	MHz		dB		ppm
01	20 to 90	+7 dBM minimum at +25°C ±2 dB from -55°C through +85°C	20	±10 ppm from -55°C to +85°C ±1 ppm from 0°C to +50°C	±0.5
02	10 to 20	0 dBM minimum at +25°C ±2 dB from -55°C through +85°C	40	±5 ppm from -55°C to +85°C ±1 ppm from -30°C to +50°C	±0.1
03	20 to 90	+7 dBM ±2 dB at +25°C ±2 dB from -55°C through +85°C	20	±10 ppm from -55°C to +85°C ±1 ppm from 0°C to +50°C	±0.5
04	10 to 20	+3 dBM ±1 dB at +25°C +2 dB from -55°C through +85°C	40	±5 ppm from -55°C to +85°C ±1 ppm from -30°C to +50°C	±0.1

Frequency aging: Measurements shall be taken at +70°C ±0.2°C every 24 hours ±1 hour, for a minimum of 30 days.

±0.1 ppm per 30 days, maximum.

±0.3 ppm per 90 days, maximum. 1/

±1 ppm per year, maximum. 1/

Terminal strength: MIL-STD-202, method 211, test conditions A and B.

Applied force: 5 pounds.

Bend: Five at 45° each.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-O-55310 and MIL-STD-202, method 204.

Nonoperating: Test condition B.

Operating: Test condition C.

1/ This is a performance requirement of the oscillator. Support data shall be presented showing that this requirement shall be met.

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Ambient pressure:

Nonoperating: In accordance with MIL-O-55310.

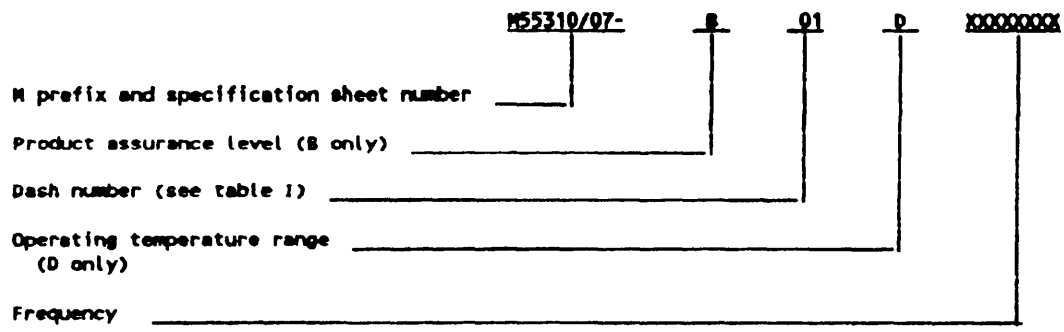
Operating: MIL-STD-202, method 105, test condition B.

QUALITY ASSURANCE PROVISIONS

Group C inspection: Not applicable.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and code alphas, and numeric number. See example:

EXAMPLE



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0677-04)

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